

COSE361: Artificial Intelligence Assignment #2

1.1 Multiple Linear Regression

1. Load the given codebase (assignment2_multi-regression.py)
 - A. Load the given dataset
 - B. Do not change the dataset and its split
2. Implement Multiple Linear Regression (using scikit-learn):
 - A. Train the model using scikit-learn to find model parameters $(\theta_0, \theta_1, \theta_2, \theta_3, \theta_4)$
 - B. Calculate MSE using the predictions from the regression model and the actual target values on the testing set
3. Implement Multiple Linear Regression & gradient descent (from scratch):
 - A. Do not use scikit-learn for regression (NumPy and standard Python are allowed)
 - B. Train the model based on the gradient descent algorithm to find model parameters $(\theta_0, \theta_1, \theta_2, \theta_3, \theta_4)$ (Choose your own learning rate)
 - C. Calculate MSE using the predictions from the regression model and the actual target values on the testing set

[Ref1] https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

1.2 Decision Tree

1. Load the given codebase (assignment2_decision-tree.py)
 - A. Load the given dataset (Breast Cancer dataset)
 - B. Do not change the dataset and its split
2. Implement Decision Tree Classifier (using scikit-learn):
 - A. Train the model using scikit-learn with criterion='gini', max_depth=3, random_state=42
 - B. Calculate accuracy on the testing set
 - C. Visualize the built tree
3. Implement Decision Tree Classifier (from scratch):
 - A. Do not use scikit-learn for decision tree (NumPy and standard Python are allowed)
 - B. Train the model with criterion='gini', max_depth=3
 - C. Calculate accuracy on the testing set
 - D. Visualize the built tree (you can simply write down the list of nodes using indent rather than graphical visualization)

[Ref2] <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>

1.3 Principal Component Analysis (PCA)

1. Load the given codebase (assignment2_PCA.py)
 - A. Load the given dataset (Iris dataset)
 - B. Do not change the dataset
 - C. Use the features (X) and the target (y) for visualization purposes.
2. Scale the features to have zero mean and unit variance
3. Implement PCA (using scikit-learn)
 - A. Apply PCA to reduce the dimensionality of the standardized Iris data to 2 principal components
 - B. Calculate the explained variance ratio for each of the selected components and the total explained variance by these 2 components.
 - C. Visualize a scatter plot of the 2D PCA-transformed data. Color the points in the scatter plot according to their original Iris species (the target labels)
4. Implement PCA (from scratch)
 - A. Do not use scikit-learn for PCA (NumPy and standard Python are allowed)
 - B. Implement PCA and apply PCA to reduce the dimensionality of the standardized Iris data to 2 principal components
 - C. Calculate the explained variance ratio for each of the selected components and the total explained variance by these 2 components.
 - D. Visualize a scatter plot of the 2D PCA-transformed data. Color the points in the scatter plot according to their original Iris species (the target labels)

[Ref3] <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

1.4 Convolutional Neural Networks (CNNs)

1. Load the given codebase (assignment2_CNN.py)
 - A. Load the given dataset (CIFAR-10 dataset)
 - B. Do not change the dataset and its split (augmentation on the given train data is okay)
2. Achieve 65% test accuracy on the given dataset by adjusting the configuration (network structure, learning rate, etc)
 - A. Do not exceed 30 training epochs

2. Submission

Submit a single ZIP file (StudentID_Name.zip) containing:

1. **Python code files (.py) (.ipynb files are not allowed)**
2. **Report (.pdf)**
 - A. (Task 1.1) Multiple Linear Regression**
 - i. Present the parts of the code you wrote with brief explanations
 - ii. Present the <screenshot> for the learned parameters and MSE from both library and scratch implementations.
 - iii. Brief discussion on the results
 - B. (Task 1.2) Decision Tree**
 - i. Present the parts of the code you wrote with brief explanations
 - ii. Present the <screenshots> for accuracy results and tree visualization from both library and scratch implementations.
 - iii. Brief discussion on the results
 - C. (Task 1.3) Principal Component Analysis (PCA)**
 - i. Present the parts of the code you wrote with brief explanations
 - ii. Present the <screenshots> for explained variance ratio results (each, total for PC1&2) and visualizations from both library and scratch implementations.
 - iii. Brief discussion on the results
 - D. (Task 1.4) Convolutional Neural Networks (CNNs)**
 - i. Present the parts of the code you wrote with brief explanations
 - ii. Present the <screenshot> for training logs + results
 - iii. Brief discussion on the results